

A Mathematical Framework for Contractive Computational Dynamics with a Thermodynamic Bridge

Local Signal Exchange, Information-Theoretic Mismatch, and Failure Conditions

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July 2026

Abstract

The deposited paper *Micah's New Law of Thermodynamics* proposed that local signal exchanges progressively dissipate differences and thereby connect equilibration, synchronization, information processing, cognition, and gravity. The original paper contained a useful local-to-global intuition but made two claims that its mathematics did not establish: that arbitrary interacting systems must erase differences, and that decreasing disagreement is generically identical to increasing thermodynamic entropy.

This revision replaces that universal law with a typed **mismatch-dissipation principle**. The principle is not an additional law of thermodynamics. It is an operational rule for constructing and testing models: an application must declare a state space, a nonnegative discrepancy or energy, the local dynamics, invariants or stationary references, a nonnegative dissipation term, external inputs, observables, and failure boundaries. The central object is a balance equation of the form $\text{change in mismatch} = -\text{internal dissipation} + \text{supplied mismatch power}$.

Six mathematical families make the distinction concrete. Symmetric consensus contracts disagreement under a spectral condition. Convex graph-gradient flows dissipate an edge potential. Forced systems obey an input-output balance and may retain residual mismatch. Reversible finite Markov dynamics dissipate relative entropy; when the stationary law is Gibbs and the rates have a justified thermodynamic interpretation, this yields a genuine free-energy and entropy-production bridge. Identical symmetric phase oscillators dissipate a phase potential only within a model-specific regime. Damped waves dissipate mechanical energy under declared boundary conditions without necessarily becoming spatially uniform.

An operational safety question exposes the same bookkeeping problem. If failures become rarer while deployment exposure and worst-case consequences grow, incident frequency alone cannot establish that risk declined. The extended framework therefore pairs event frequency with exposure and a severity distribution, then keeps any physical energy or declared informational storage balance separate from behavioral harm. This makes rare-event claims testable without treating unintended consequences as a conserved substance.

The component results have substantial prior art. The contribution of this revision is a corrected synthesis, a common certificate, an executable boundary suite, and a claim-typing rule that prevents consensus variance, information divergence, free energy, phase potential, mechanical energy, heat, and thermodynamic entropy from being treated as interchangeable. The proposed neuroscience and gravity applications remain separate research programs that require independent equations, measurements, and falsifiers.

Scope and evidence boundary

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16922506

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1. From a Universal Law to a Testable Principle

The 2025 deposit described equilibrium as a distributed computation [12]. Neighboring components exchange partial information about their differences; repeated local updates produce a global macroscopic state. This is a valuable organizing idea. It appears in consensus algorithms, diffusion, gradient flows, Markov relaxation, oscillator synchronization, damped waves, control, inference, and learning [1-8,11].

The phrase *appears in* is not the same as *proves one universal mechanism*. Interactions can be contractive, conservative, periodic, expansive, frustrated, chaotic, driven, or disconnected. A local exchange can lower one functional while raising another. A system can dissipate mechanical energy while maintaining a spatial pattern, or maintain a nonequilibrium current while producing entropy. A controller can lower an error by spending external energy.

The corrected principle is therefore methodological:

Mismatch-Dissipation Principle. A claimed mismatch-dissipation process is scientifically specified only when a model derives a nonnegative discrepancy functional and an explicit balance showing how declared local dynamics reduce it, preserve it, increase it, or offset its reduction through inputs. The word *informational* is reserved for cases in which the functional is an information divergence or is connected to one by a derived equation.

The principle does not say every real system satisfies such a balance with a strictly positive dissipation term. It says what must be exhibited before language such as *dissipation*, *equilibration*, or *information loss* is promoted from analogy to a mathematical claim.

2. The Dissipation Certificate

Let the application certificate be

$$\mathcal{C} = (\mathcal{X}, z, \mathcal{D}, F \text{ or } f, \mathcal{I}, \mathcal{R}, \mathcal{U}, \mathcal{O}, \mathcal{B}), \quad (1)$$

where:

- $\mathcal{X}$ is the state space, geometry, and boundary conditions;
- z is the state, including units;
- $\mathcal{D}(z) \geq 0$ is a discrepancy, potential, divergence, or energy;
- F or f is the discrete update or continuous vector field;
- $\mathcal{I}$ is the invariant, stationary distribution, or reference set;
- $\mathcal{R}(z) \geq 0$ is the internally generated dissipation rate;
- $\mathcal{U}$ contains forcing, controls, reservoirs, and boundary fluxes;
- $\mathcal{O}$ identifies measurable observables and uncertainty; and
- $\mathcal{B}$ lists the assumptions and failure boundary.

For continuous time, the desired certificate has the form

$$\frac{d}{dt}\mathcal{D}(z_t) = -\mathcal{R}(z_t) + \mathcal{P}_{\mathcal{U}}(z_t, u_t), \quad \mathcal{R} \geq 0, \quad (2)$$

and in discrete time it has the form

$$\mathcal{D}(z_{k+1}) - \mathcal{D}(z_k) \leq -\mathcal{R}_k + \mathcal{P}_k. \quad (3)$$

The input-power term is not a nuisance to be omitted. It distinguishes an autonomous relaxation from an open, controlled, or driven system. A strict descent theorem requires that the supplied term vanish or be dominated by the internal dissipation. A steady residual can be the correct predicted outcome.

The same notation must not erase physical distinctions. Depending on the certificate, D may be squared disagreement, an optimization objective, relative entropy, Gibbs free-energy excess, a phase potential, or mechanical energy. Only a thermodynamic specialization may call R/T an entropy-production rate, and only after temperature, reservoirs, probabilities, and balance laws have been supplied.

3. Certificate I: Linear Consensus

Let x_k in $\mathbb{R}^n$, let $\mathbf{1}$ be the all-ones vector, and define

$$\begin{aligned}\bar{x} &= \frac{1}{n} \mathbf{1}^\top x, \\ P &= I - \frac{1}{n} \mathbf{1} \mathbf{1}^\top, \\ V(x) &= \frac{1}{n} \|Px\|_2^2.\end{aligned}\tag{4}$$

Consider

$$x_{k+1} = Wx_k.\tag{5}$$

If W is real, symmetric, and doubly stochastic, the arithmetic mean is invariant. If its operator norm on the disagreement subspace satisfies

$$\rho = \|W|_{\mathbf{1}^\perp}\|_2 < 1,\tag{6}$$

then

$$\begin{aligned}V(x_k) &\leq \rho^{2k} V(x_0), \\ x_k &\longrightarrow \bar{x}_0 \mathbf{1}.\end{aligned}\tag{7}$$

The proof follows by mean preservation, invariance of $\mathbf{1}^\perp$, and $Px_k = W^k Px_0$. This is a consensus theorem, not a thermodynamic law [1,2].

For an undirected weighted graph with Laplacian L ,

$$\begin{aligned}\dot{x} &= -Lx, \\ \dot{V} &= -\frac{2}{n} (Px)^\top L(Px) \leq 0.\end{aligned}\tag{8}$$

Connectedness makes the null space one-dimensional and yields strict decay away from consensus. A disconnected graph retains differences between component means. A balanced directed graph may preserve a weighted or arithmetic mean, but a generic row-stochastic update need not. Negative anti-diffusive weights can reverse the sign. These are boundary conditions in the certificate, not minor caveats.

4. Certificate II: Nonlinear Convex Graph-Gradient Flow

The original local-exchange intuition is not limited to quadratic variance. For an undirected graph with nonnegative weights a_{ij} , choose a differentiable even edge potential ϕ and define

$$\Phi(x) = \sum_{(i,j) \in E} a_{ij} \phi(x_i - x_j). \quad (9)$$

Translation invariance gives $1^\top \text{grad } \Phi = 0$. Let $M(x)$ be symmetric positive semidefinite on the disagreement subspace and consider

$$\dot{x} = -M(x) \nabla \Phi(x). \quad (10)$$

The potential has the exact balance

$$\dot{\Phi} = -\nabla \Phi(x)^\top M(x) \nabla \Phi(x) \leq 0. \quad (11)$$

This is the clean mathematical form of local mismatch dissipation for a broad gradient-flow family. It still does not guarantee global consensus. The critical set may contain multiple attractors; the graph may be disconnected; the potential may not be convex on the quotient space; or mobility may vanish.

Suppose M is bounded below by $m > 0$ on the relevant quotient space and Φ satisfies a Polyak-Lojasiewicz inequality

$$\frac{1}{2} \|\nabla \Phi(x)\|_2^2 \geq \mu [\Phi(x) - \Phi_*]. \quad (12)$$

Then

$$\Phi(x_t) - \Phi_* \leq e^{-2\mu t} [\Phi(x_0) - \Phi_*]. \quad (13)$$

The rate is thus a derived consequence of geometry, mobility, and the chosen potential. Calling every interaction “dissipative” would discard the very conditions that make the result true [2].

5. Certificate III: Forced and Open Systems

Let the gradient system receive an input through B :

$$\dot{x} = -M(x) \nabla \Phi(x) + Bu(t). \quad (14)$$

Its balance is

$$\dot{\Phi} = -\nabla \Phi^\top M \nabla \Phi + \nabla \Phi^\top Bu. \quad (15)$$

The second term can sustain or increase mismatch. If $M \succeq mI$ on the active subspace, Young’s inequality gives

$$\dot{\Phi} \leq -\frac{m}{2} \|\nabla \Phi\|_2^2 + \frac{1}{2m} \|Bu\|_2^2. \quad (16)$$

Combining this with Equation (12) yields an input-to-state mismatch bound,

$$\Phi(t) - \Phi_* \leq e^{-m\mu t} [\Phi(0) - \Phi_*] + \frac{1}{2m} \int_0^t e^{-m\mu(t-s)} \|Bu(s)\|_2^2 ds. \quad (17)$$

Equation (17) replaces the binary statement “the system converges or fails” with a quantitative prediction. A bounded input produces a bounded residual. A controller may lower mismatch by supplying work. A living system may maintain organized nonequilibrium structure through continuing metabolic throughput. None of these cases contradicts internal dissipation; they require the input term to remain visible.

6. Certificate IV: Relative Entropy and a Legitimate Thermodynamic Bridge

The word *informational* becomes literal for probability dynamics. Consider a finite continuous-time Markov chain with rates k_{ij} , probability $p_i(t)$, and

$$\dot{p}_i = \sum_j [p_j k_{ji} - p_i k_{ij}]. \quad (18)$$

Let $p_i > 0$ be stationary and suppose detailed balance holds:

$$\pi_i k_{ij} = \pi_j k_{ji} = c_{ij}. \quad (19)$$

Define the relative entropy and density ratio

$$D_{\text{KL}}(p||\pi) = \sum_i p_i \ln \frac{p_i}{\pi_i}, \quad r_i = \frac{p_i}{\pi_i}. \quad (20)$$

Direct differentiation gives

$$\frac{d}{dt} D_{\text{KL}}(p||\pi) = -\frac{1}{2} \sum_{i,j} c_{ij} (r_i - r_j) [\ln r_i - \ln r_j] \leq 0, \quad (21)$$

because the logarithm is monotone. This is an information-dissipation theorem for the declared Markov process [3,4]. Reversibility, irreducibility, and the chosen reference distribution are real assumptions.

A thermodynamic interpretation requires more. If states have energies E_i and the stationary law is Gibbs at inverse temperature β ,

$$\pi_i = \frac{e^{-\beta E_i}}{Z}, \quad Z = \sum_i e^{-\beta E_i}, \quad (22)$$

define the nonequilibrium free energy

$$\mathcal{F}[p] = \sum_i p_i E_i + k_B T \sum_i p_i \ln p_i. \quad (23)$$

Then

$$\mathcal{F}[p] - \mathcal{F}[\pi] = k_B T D_{\text{KL}}(p||\pi), \quad (24)$$

and for this fixed detailed-balance relaxation,

$$\dot{S}_{\text{tot}} = -k_B \frac{d}{dt} D_{\text{KL}}(p||\pi) \geq 0. \quad (25)$$

Equations (21)-(25) supply the bridge absent from the deposited paper. The decreasing quantity is relative entropy or free-energy excess; the total entropy production has the opposite sign. Heat, work, system entropy, and medium entropy are not created by renaming a variance. They arise from the stochastic and thermodynamic model [5,6,9,10].

Driven Markov chains may have stationary currents and positive housekeeping entropy production. Nonreversible chains can still contract suitable divergences, but the simple detailed-balance identity changes. These cases belong in an expanded certificate, not under a universal equilibrium slogan.

7. Certificate V: Phase Oscillators

For identical natural frequencies viewed in a rotating frame, a symmetric Kuramoto network can be written [7,8]

$$\dot{\theta}_i = -K \sum_j a_{ij} \sin(\theta_i - \theta_j). \quad (26)$$

Define the phase potential

$$U(\theta) = K \sum_{(i,j) \in E} a_{ij} [1 - \cos(\theta_i - \theta_j)]. \quad (27)$$

Equation (26) is $\dot{\theta}_i = -\nabla U$, hence

$$\dot{U} = -\|\nabla U\|_2^2 \leq 0. \quad (28)$$

This is a valid mismatch-dissipation certificate for a declared phase model. It is not yet a global synchronization theorem. The phase space is a torus; multiple critical points and winding states can exist. Convexity is local to a phase-cohesive region, commonly represented by edge differences below $\pi/2$. Connectivity and coupling strength matter.

With heterogeneous natural frequencies ω ,

$$\begin{aligned} \dot{\theta} &= \omega - \nabla U, \\ \dot{U} &= -\|\nabla U\|_2^2 + \nabla U^T \omega. \end{aligned} \quad (29)$$

The frequency term acts like an input and can reverse instantaneous descent. Frequency locking, phase locking, coherence, and thermodynamic equilibrium are different observables. In neuroscience, a Kuramoto order parameter is a population phase statistic; it is not automatically heat, entropy, perceptual binding, or consciousness.

8. Certificate VI: Damped Waves

To preserve the wave-based core of the historical paper, consider a field $q(x, t)$ on a domain Ω with periodic, fixed, or no-flux boundary conditions:

$$\partial_{tt}q + \gamma \partial_t q - c^2 \Delta q + V'(q) = 0, \quad \gamma \geq 0. \quad (30)$$

Its mechanical energy is

$$E[q] = \int_{\Omega} \left[\frac{1}{2} (\partial_t q)^2 + \frac{c^2}{2} \|\nabla q\|^2 + V(q) \right] dx. \quad (31)$$

Integration by parts under the declared boundary conditions yields

$$\frac{dE}{dt} = -\gamma \int_{\Omega} (\partial_t q)^2 dx \leq 0. \quad (32)$$

Equation (32) is genuine wave-energy dissipation. It does not imply that the field value becomes uniform. The potential can support multiple minima, topological structures, standing configurations, or pinned boundaries. With $\gamma=0$, energy is conserved. With negative damping, the system is active and energy grows. With external forcing or open boundaries, power and flux terms must be added.

The phrase *phase-wave differential* can therefore be made mathematically specific only after selecting a field, phase variable, dispersion relation, damping mechanism, and measured energy or divergence. “Wave-like” alone is not a dissipation certificate.

9. The Cross-Model Map

The six certificates expose both a common architecture and irreducible differences:

Family	State	Decreasing object	Required bridge	Typical failure
linear consensus	real node values	squared disagreement	none to network dynamics	disconnected, periodic, unbalanced, negative weights
graph gradient flow	node values on a quotient space	edge potential	potential and mobility	nonconvex critical sets, vanishing mobility
forced system	state plus input	residual balance	input channel and norm	supplied power exceeds dissipation
reversible Markov chain	probability simplex	relative entropy / free-energy excess	stationary law; detailed balance	driving, nonreversibility, reducibility
phase oscillator	torus phases	phase potential in a declared regime	coupling graph and phase geometry	heterogeneity, frustration, winding states
damped wave	field plus velocity	mechanical energy	damping and boundary conditions	no/negative damping, forcing, open flux

The shared narrative is local transformation of a distributed mismatch under a constraint. The mathematical object being transformed is not the same in every row. That distinction strengthens rather than weakens the original insight: it turns a broad metaphor into a research program whose applications can pass or fail independently.

10. Scale, Rare-Event Severity, and Contractive Control

10.1 The ordinary problem and established safety account

Consider a deployed system that once causes one harmful unintended outcome in every thousand opportunities. A later system may fail only once in ten thousand opportunities, yet be used a hundred times more often or gain access to actions with much larger consequences. The lower per-opportunity incident rate is good evidence about one quantity, but it does not determine total incidents or total risk.

Established AI-safety work already treats accidents as unintended harmful behavior arising from such problems as side effects, reward hacking, inadequate supervision, unsafe exploration, and distribution shift [20]. For an observation window with N opportunities, unintended-event probability p_u , and nonnegative severity S , the first ordinary bookkeeping quantity is

$$H = N p_u \mathbb{E}[S \mid \text{unintended}]. \quad (33)$$

Equation (33) is exposure-adjusted expected harm, not physical energy. It also does not describe a catastrophic tail by itself. Report high quantiles, the probability of crossing a declared catastrophic threshold, and expected shortfall. One standard tail functional is conditional value at risk,

$$\text{CVaR}_\alpha(S) = \inf_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{1 - \alpha} \mathbb{E}[(S - \eta)_+] \right\}, \quad (34)$$

$$0 < \alpha < 1,$$

with the loss convention and confidence level declared [19]. Frequency, exposure, mean severity, tail severity, and detection probability are distinct observables.

10.2 Reachable variety and the control operation

As capability grows, a system may acquire more tools, longer planning horizons, more environmental interfaces, and more distinguishable behavior traces. A controller must still recognize relevant situations, select an adequate response, and keep the probability and severity of unsafe outcomes contracting despite that expanding opportunity set. In plain language: safety improves only when the regulator's effective distinctions and interventions improve faster than consequential opportunities expand.

After that operation is explicit, call the comparison the **state-space expansion versus contractive-control problem**. If P indexes a declared scale or capability level, distinguish the number of distinguishable unsafe modes $|U(P)|$ from their total probability mass $p_U(P) = \sum_{u \in U(P)} p(u|P)$. A scale-aware risk functional is

$$R(P) = N(P) \sum_{u \in U(P)} p(u|P) S(u, P). \quad (35)$$

The support can expand while unsafe probability mass contracts; neither fact alone settles Equation (35). Ashby's law of requisite variety gives the directly relevant control boundary: effective regulation must retain enough response variety to constrain the disturbances and states that matter [16]. Kolmogorov's algorithmic account motivates the narrower possibility that some individual traces lack short descriptions [17]. Rice's theorem rules out a universal decision procedure for every nontrivial extensional semantic

property of arbitrary programs [18]. Neither result proves that a particular finite model is unpredictable, that failure must rise with parameter count, or that testing and formal verification are futile.

The discriminator is empirical and operational: estimate how exposure, unsafe probability mass, consequence severity, and controller coverage change across capability levels. A claim of improved contractive control fails if the measured risk or its declared catastrophic tail rises, if relevant modes are systematically missed, or if apparent contraction disappears after exposure and detection are held comparable.

10.3 Rare-event frequency, severity, and a typed balance

Let $N_u(T)$ count detected unintended events in a window of duration T , let $\lambda = E[N_u(T)]/T$, and let M be a declared nonnegative magnitude. The measured unintended-channel flux is

$$J_u = \lambda \mathbb{E}[M]. \quad (36)$$

A decrease in λ does not establish improvement when mean magnitude or the tail rises enough to offset it. If both frequency and severity decline, J_u declines. That observation still does not identify where physical energy went, because M may be dollars, injuries, service disruption, policy violations, or another harm score.

Only after a physical or declared informational storage variable has been identified may an application write

$$\frac{dE_s}{dt} = P_{\text{in}} - P_{\text{intended}} - P_{\text{diss}} - P_{\text{boundary}} - J_u. \quad (37)$$

Every term in Equation (37) must share calibrated units. When J_u is only a behavioral-risk measure, it does not belong in this energy equation. When it is a measured energy or mismatch flux, a lower event rate without larger events can be balanced by increased intended work, dissipation, boundary export, storage, an unmeasured channel, or reduced input. Only independently bounded input, storage, intended work, and missing channels support a narrower inference about dissipation or export. In a stationary bounded-storage regime, a sustained fall in J_u must eventually be balanced elsewhere.

This preserves the type boundary already used throughout the paper. Behavioral error, algorithmic complexity, Shannon information, model uncertainty, mechanical energy, free energy, heat, and thermodynamic entropy are not synonyms. An intelligent controller can maintain local order only as an open physical system that consumes free energy and exports entropy; a behavioral incident count does not by itself establish that thermodynamic bridge.

10.4 Self-reinforcing failure dynamics

A recurrent learner can enter a subpolicy, representation, prompt pattern, tool loop, or environment-coupled feedback structure that persists and amplifies itself even though no engineer installed it as a named module. Training data, optimization, architecture, external inputs, and deployment feedback can all be causal ingredients. Learned optimization, including the possibility that a trained model itself performs optimization toward an objective that differs from the training loss, is an established adjacent concern [21].

After the causal operation is stated, call a persistent harmful case an **endogenous adversarial attractor**. The name does not imply spontaneous creation, agency, a literal virus, or conserved unintended-consequence energy. It denotes a candidate reachable failure mode. The construct earns scientific content only if an

application measures basin conditions, persistence after the trigger is removed, reproduction or propagation, resistance to ordinary correction, and consequence severity. Failure to distinguish those signatures from a transient prompt response, ordinary distribution shift, or a designed module defeats the attractor claim.

10.5 Required discriminators and scope boundary

A rare-event application of the certificate must therefore:

1. define opportunity count, detection process, and exposure;
2. report event frequency, mean severity, tail quantiles, expected shortfall, catastrophic-threshold probability, and maximum-event sensitivity;
3. test censoring, missed detection, and changing observation windows;
4. declare whether magnitude is harm, mismatch, information, or physical energy, with units and calibration;
5. verify stationarity or model storage and all material input / output channels;
6. compare unsafe-mode growth with controller coverage and response variety; and
7. perturb suspected attractors and state a result that would reject the model.

The defensible synthesis is a competition between expanding reachable variety and contractive control, not a conservation law for unintended consequences. This paper does not propose a thalamocortical-inspired governor, protected causal bottleneck, or evolutionary redesign architecture. Those are separate architectural hypotheses requiring their own manuscript and evidence.

11. Relationship to SAN Neuroscience

Neural recordings contain distributed timing, amplitude, and state changes, yet a stable neural computation need not collapse every element into one globally synchronous value. The scientific problem is to explain how local neural populations receive and transform signals while preserving task-relevant phase relations, cluster structure, or dispersion. Established accounts include phase coding, communication through coherence, predictive coding, and coupled-oscillator dynamics [7,8,11].

A paper-specific operation would identify what local signal is received, how relative timing or another declared state changes, which downstream population receives the transformed pattern, and how returned sensory or internal evidence updates it. After that receive-transform-project-and-re-sample operation is understandable, **Self Aware Networks (SAN)** names the broader proposed program. Within it, **Phase Wave Differentials, Neural Array Projection Oscillation Tomography (NAPOT)**, neural rendering, coherence, and splay-state dispersion are candidate neural objects for a dissipation certificate, not results selected by the present mathematics [13]. Each earns application-specific status only if recordings or perturbations distinguish its predicted transformation from the established alternatives.

A SAN specialization would have to declare, at minimum:

1. whether the state is spike phase, membrane voltage, dendritic current, synaptic efficacy, field potential, or a multiscale combination;
2. whether the geometry is Euclidean, circular, manifold-valued, or a graph of local neural arrays;
3. the exact mismatch functional and its units;
4. the local transformation rule, delays, noise, excitation, and inhibition;
5. an input / output balance for sensory and motor forcing;
6. observables measurable by spikes, intracellular recording, LFP, EEG, MEG, calcium imaging, or perturbation; and
7. a result that would distinguish the SAN operator from generic predictive coding, communication-through-coherence, or oscillator synchronization.

The framework accommodates both coherence and structured phase dispersion. Synchronization need not mean a global all-on state. A network can preserve relative phase offsets, cluster structure, or splay states, and a perturbation can reorganize those relations. What matters is the declared functional and balance, not a presumption that coherence is success and decoherence is failure.

Consciousness is not a consequence of Equations (7), (11), (21), (28), or (32). Those equations can become components of a SAN model only after the receive- transform-project architecture and its measurements are independently supplied.

12. Relationship to SIT, QGTCD, and Gravity

A proposed relaxation field near matter would have to predict a measurable clock, trajectory, lensing, or energy-flow difference after standard gravitational and material effects are controlled. General relativity and ordinary nonequilibrium thermodynamics are the established accounts to beat; a network analogy is not a metric theory. The required plain-language operation is to declare the field and source, specify how they change local dynamics, track energy and stress, and derive a discriminator with consistent units and standard limits.

Only after that operation is stated do **Super Information Theory (SIT)** and **Quantum Gradient Time Crystal Dilation (QGTCD)** name the author's proposed physics programs. The original paper linked signal dissipation to local time density and gravity [12]. The present framework does not derive those programs. A physics specialization requires a covariant action or other well-defined dynamics, dimensional consistency, a stress-energy account, conservation identities, standard limits, parameter constraints, and observations.

The certificate remains useful as a design constraint. If a proposed time- density or coherence field relaxes, the paper should identify its potential, kinetic term, damping or open-system origin, source, and energy flow. If the field is conservative, calling the dynamics dissipative is wrong. If damping is inserted phenomenologically, the covariant completion and destination of the energy must be stated. A network consensus theorem cannot supply those missing physics steps.

SIT and QGTCD should therefore cite this paper for the typed mismatch-balance method, not as proof that gravity is thermodynamic dissipation [14,15].

13. Prior Art and What This Revision Adds

Consensus convergence, convex Lyapunov functions, Markov-chain entropy gradient flows, stochastic entropy production, Kuramoto synchronization, and damped-wave energy estimates all predate this revision. In particular, Olfati-Saber and Murray formalized graph conditions for consensus [1]; Mangesius and Delvenne related linear consensus to convex Lyapunov and gradient-flow structures [2]; Maas constructed an entropy-gradient-flow geometry for reversible finite Markov chains [3]; Schnakenberg and Seifert developed network and trajectory accounts of entropy production [4,5]; and Dörfler, Chertkov, and Bullo derived network-dependent synchronization conditions [7].

The revision does not claim those component theorems as new. Its contribution is the correction and operational synthesis:

- one typed certificate across several mechanisms;
- an explicit rule limiting the word *informational*;
- a derived thermodynamic bridge rather than a variance / entropy analogy;
- input terms that keep open and living systems in scope without pretending they are isolated equilibrating systems;

- executable identity checks and adversarial boundary cases; and
- a cross-domain promotion rule requiring equations, units, observables, and falsifiers.

This contribution can be judged on clarity, usefulness, and reproducibility without requiring the reader to accept a new fundamental law.

14. Reproducibility and Falsification

The companion v2 runner evaluates algebraic checkpoints for all six families. It verifies identities, not merely decreasing plots. The current fixed packet contains no fitting, randomness, or external data. Its passing output means only that the declared finite calculations match the equations.

Claim	Required evidence	Direct failure
consensus mismatch contracts	invariant plus spectral / ergodic condition	admitted trajectory with expansion or a missing mode
graph potential dissipates	gradient-flow identity and positive mobility	nonsymmetric / active term reverses the balance
input produces bounded residual	gain and input norm bound	residual exceeds the proved bound
KL divergence dissipates	stationary reference and rate model	derivative or detailed-balance identity fails
thermodynamic bridge holds	Gibbs state, temperature, heat / work definitions, balance	incompatible units or missing medium entropy
phase potential decreases	identical / compensated frequencies and symmetric coupling	input term makes $U_{\dot{}} > 0$
wave energy dissipates	positive damping and controlled boundary flux	zero / negative damping or unaccounted forcing

Independent network-dynamics and stochastic-thermodynamics review remain open. The certificate is falsifiable in each application because every claim names the quantity, assumptions, and sign test that can fail.

15. Conclusion

The historical paper saw a real recurring architecture: distributed components can transform local differences through repeated exchanges, and a macroscopic state can sometimes be predicted without simulating every microscopic detail. The original title and entropy claim pushed that insight beyond what the mathematics established.

The mismatch-dissipation principle preserves the research program in a stronger form. It replaces universality by a certificate, analogy by a balance, and a single overloaded word *entropy* by typed mathematical objects. Linear consensus, nonlinear gradient flows, reversible Markov relaxation, phase oscillators, damped waves, and forced systems then become comparable without becoming identical.

This is not a fourth law of thermodynamics. It is a disciplined framework for asking when local signal exchange truly dissipates a declared mismatch, when that mismatch is informational, when it has a thermodynamic interpretation, and where the conclusion fails. That framework is broad enough to guide SAN and SIT applications while requiring each application to earn its own scientific claim.

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